Investigating the Climatic Effect on Reliability Assessment of a Composite Power System Using a Bayesian Network Base on Minimal Cutset

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***Abstract*— This paper presents a novel method to evaluate the reliability of the composite power system considering the effect of climate. In recent years, Bayesian networks have become a powerful tool in modeling and logical analysis of engineering systems and power systems. The main purpose of this paper is to formulate and elaborate a suitable framework for the analytical evaluation and estimation of composite power system reliability indices, consisting of generating units and transmission lines, using the Bayesian network considering the climatic effect of the model. The Bayesian network corresponding to the composite power system is formed in the proposed method using the concept of minimal cutset. The constructed Bayesian network provides an evaluation of the loss of load probability index of the system considering the effects of weather and climatic conditions, and various other analyzes from the perspective of reliability in the desired power system. The proposed methodology is implemented on Roy Billiton test system and the results are evaluated and analyzed.**

Keywords*— Bayesian network, Composite power system, Reliability, Weather effect.

I. INTRODUCTION

Reliability in a power system is the ability of the system to supply the electricity required by subscribers at an acceptable level of continuity and durability. In terms of reliability in the power system, we deal with the concept of adequacy, which implies that the equipment needed to generate sufficient power and the transmission and distribution equipment needed to deliver power to the load points. Due to the effect of the fault mechanism of some components, particularly transmission lines, Weather condition has an essential effect on reliability indices of the power system; therefore different weather and geographical conditions can affect outage rate values . Consequently, it causes changing in different reliability indices to evaluate the adequacy of the power system.

Bayesian network (BN) is an efficient graphical model that expresses the possible relationships between large numbers of variables; hence it is an appropriate tool for different inferences in many engineering sciences. In addition to other engineering systems, Bayesian network application has been applied in power system studies. These studies include Bayesian networks for detecting fault types and damaged parts in the power system [1], investigating the effects of cascading outages and global shutdowns [2], estimating and predicting transmission lines fault due to weather conditions [3], risk analysis in management system maintenance [4] and studies of wind energy conversion systems [5].

Among the applications of the Bayesian network in the power system, the use of BN in reliability studies is an important issue but has not yet been extensive research in this regard. In some applications, the use of Bayesian networks in distribution networks has been discussed [6]. In [7], a Bayesian network method is implemented to analyze the reliability of a composite system consisting of generating units and transmission lines with an emphasis on the importance of the components of the power system. This methodology is based on the Bayesian network learning capability. It can be applied to larger power systems as well. In this reference, the constructed Bayesian network is used for various probabilistic discriminant analyzes, such as ranking the importance of different components of the system, including power plants and transmission lines from the reliability point of the system. In this method, a number of the critical components, including transmission lines having small unavailability, are eliminated. Thus the model is not entirely accurate to evaluate the reliability of the power system. Moreover, learning the BN structure in this method is based on the Monte Carlo method, which is a time-consuming and complicated process. Also, the construction of this reference model depends on forced outage rate (FOR) of system components. In order to add the weather effect model, the corresponding Bayesian network must be regenerated, while in this paper, all the components of the power system are considered, and it does not depend on (FOR) of the components to construct the BN. So the results are more accurate in this way. Also, the addition of the climate model, does not change the main Bayesian model system. In [8], utilizing minimal cutsets (MCs), BN of the composite power system is constructed. It is used for modeling common cause failures and protection system failures in the power system. None of the previous applications of BN, the effect of the weather, has not been considered in order to accurately assess the reliability of the power system.

The failure rate of the components of the system, such as transmission lines in the open environment, is a function of weather conditions. The failure rate of a transmission line under adverse climatic conditions may be much higher than their failure rate in normal weather. In order to achieve a more comprehensive model in the detailed reliability assessment of the power system, how to incorporate the geographical

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conditions, and climatic effects in the BN model have been considered.

The main purpose of this paper is to develop and elaborate a suitable framework for the analytical evaluation and estimation of composite power system reliability indices using Bayesian networks. Then, using the concept of minimal cutset (MC), we propose a Bayesian network corresponding to the composite power system. In addition to estimating the loss of load probability (LOLP) indices, it is proposed to consider the effect of weather and climatic conditions and different analysis in terms of reliability in the desired power system using a Bayesian network.

The structure of this paper will continue as follows: In section 2, we introduce the Bayesian network. In section 3, Bayesian network modeling corresponds to the composite power system. In section 4, the necessary modifications of the Bayesian network are investigated by considering the climatic effect, and the simulation results are implemented on the RBTS system. Section 5 summarizes the results of this paper.

II. BAYESIAN NETWORK

Bayesian network is an efficient graphical model that expresses the relationship and possible information between a large number of variables. A Bayesian network based on causal relationships between its variables, prior information, and the experience of experts and using learning algorithms from training samples. Besides, Bayesian networks allow for continuous and discrete variables to be considered simultaneously. In cases with a large number of variables, these relationships are determined based on learning algorithms. According to these features, Bayesian networks can also be considered as a suitable tool for assessing the reliability of the power system. Each Bayesian network has both qualitative and quantitative parts. The qualitative part (called structure or graph) consists of two parts of the node and the orientation branches called edges. In these networks, nodes represent system variables, and edges represent the logical relationship between variables. Quantitative parts or parameters include the marginal probability distributions (MPDs) or the conditional probability distributions (CPDs) that are assigned to each node. To formulate a Bayesian model for evaluating the reliability of a composite power system, two main parts, namely structure and parameters, must be identified and determined. The basic theorem in the Bayesian grid used to form the BN corresponding to the composite power system is as follows:

If it is assumed that in the directed acyclic graph (DAG), the probability distribution on the set of variables is equal to P and if <G,P> is true Markov's condition, then the probability distribution P is equal to the product of the probability distributions of each node conditionally on its parent node(1):

$$P = \prod_{i=1}^{n} (x_i | PA_{x_i}) \quad (1)$$

For example, in Fig. 1, the probability distribution between the set of variables is computed as P (B, E, R, A, C) without considering conditional independence in the form of relation (2):

$$P(B,E,R,A,C)=P(B)*P(E)*P(R|E)*P(A|R,E)*P(C|A) \quad (2)$$

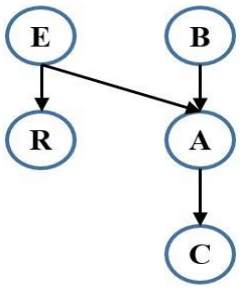


Fig. 1. An example of a Bayesian network and computing the probability distribution of its variables

III. CONSTRUCTING A BAYESIAN MODEL TO EVALUATE THE RELIABILITY OF THE POWER SYSTEM

In this section, the method of extracting minimal cutsets in the composite power system will be presented. Then the corresponding BNs based on MC will be constituted. This work has been used in the security assessment of power system, taking into account the constant load level to determine the critical point of the system and modeling the protection system failure in reference [8].

A. Construction Bayesian model based on MC

In [9,10], a method is expressed to determine MCs for engineering systems and power systems. In these methods, the power system is considered as a typical graph, and we cannot consider power system constraints. So, these methods will not work properly in the power system. Here, by considering the necessary modifications to the method presented in reference [8], the MCs of the RBTS test system are determined, and the corresponding Bayesian network is formed.

In a power system, an MC is a set of generating or transmitting network equipment that, in case of failure, causes not to supply the system or violates the power system constraints.

The first step in extracting system MCs is to determine the maximum order of the cutsets. Order of the cutsets is the number of elements in a system that must be disabled to cause system failure, which is represented here by M. After specifying M, the second step is to obtain a matrix that represents all states containing the outage of one to M of the component and is called the structure matrix (SM).

For an arbitrary system with N binary members, the structure matrix dimensions will be N × Q, which Q is calculated from (3).

$$Q=\sum_{i=1}^{M}\binom{N}{i} \quad (3)$$

The x_{ij} is the element of the SM matrix, which represents the state of the element I, in the state of j of the above system states. x_{ij} is a binary element that has two values of one or zero. The value of one represents the state of the fault, and the value of zero represents the normal state of element ith. Columns represent the different states of the system consisting of one to M elements outage. It is also a priority in the formation of each column with the extracted elements (with value 1) so that if there is a cutset with the different elements being extracted, there is no need to continue exploring that state in other columns of SM. Therefore rows 1 to N include all outage state of one component. Similarly,

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columns $1+\sum_{i=1}^{k-1}\binom{N}{i}$ to $\sum_{i}^{k}\binom{N}{i}$ for ($1 \leqslant k \leqslant M$) represent all elements of the k outage of the system components.

In the third step, DC or AC optimal power flow can be used to extract MCs in the composite power system. In this study, focusing on the concept of active power generation and consumption in reliability issues, the DC optimal power flow is used, which is presented in the set of relationships (4).

$$\text{Minimize, } \lambda \tag{4}$$

s.t

$$\sum_{k=1}^{NB} a_{il} P_l \leq CG_i \quad i \in \beta_B$$

$$CT_{min} \leq P_L \leq CT_{max} \quad L \in \beta_T$$

$$(L_{pk} - P_k)/L_{pk} \leq \lambda \quad k \in \beta_l$$

DC-OPF in the set of equations (4) is used for simultaneous failure analysis of the selected components, and the amount of power not supplied is calculated in i_{th} load point, and objective function (λ) is calculated. L_{pk} is peak load at load point k and P_k is amount of arrival power at load point k. a_{il} represents elements of nodes-branches incidence matrix, and P_l is control variable indicating the amount of effective power flow of branch L. If the demand for a load point is not supplied in the case under investigation, we have a positive value for (λ), and the MC is then specified.

After determining the order of the system and forming the SM matrix, which contains all M-order outages of the system, the various states of the system are evaluated using the above DC-OPF, and the amount power value which reaches each load point is calculated. If the value of the demand is higher than the amount of arrival power for ith load point, this state is considered as a cutset for the system.

SM_{new} is a matrix with dimensions $N \times (Q+1)$.indeed, SM_{new}consists of SM matrix with added row obtained from evaluation all state of SM matrix. After evaluation of the j_{th} state with DC-OPF program, the EF_j is equal to one if the arrival power of one or more load points is lower than the peak load in the same load point. Such a condition is considered a cutset for the power system. According to Definition MC, to reduce the number of evaluations of the SM matrix modes, if EF_j is equal to 1, the other columns after the j_{th} column containing those elements are omitted from the SM matrix. Therefore, the proposed method is applicable to larger systems with more elements. The block diagram of the MC extraction method for the composite power system is shown in Fig. 2.

B. BN corresponding to RBTS network in composite power system

In the first step, we determine the RBTS network MCs at the peak load level, which is the worst operating condition by the method described in Section III.A. To do this, we assigned order M=3. Then the SF matrix is formed. SF matrix will consider all the fault states to the M order of the RBTS system. The RBTS system has 11 generation units and 9 transmission lines; therefore the dimension of SF corresponding to it is 20×1350.

In the next step, each of the SF matrix columns, which

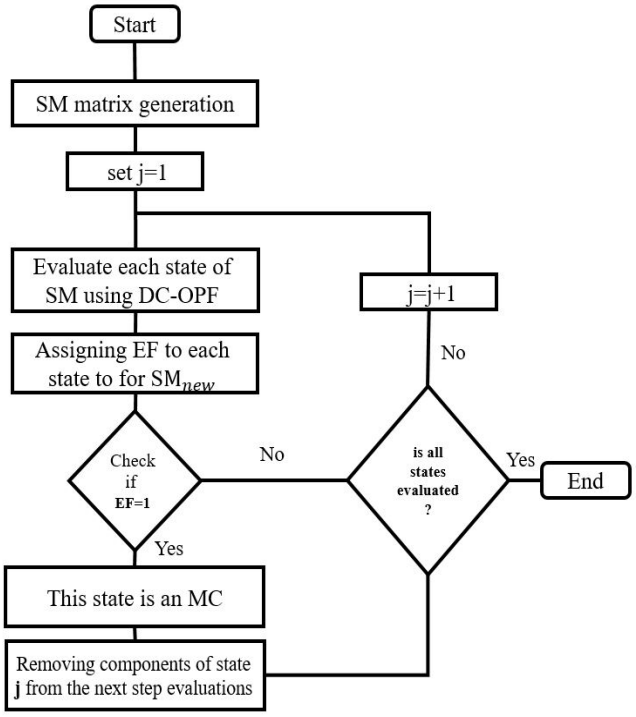


Fig. 2. Flowchart of MC determination algorithm

represents a different state of the component outages, evaluate using DC-OPF program to determine which state is an MC. For the peak load, we shall have 57 MCs consisting of one first-order MC, 33 second-order MCs, and 23 third-order MCs and are listed in Table I.

To form a BN, follow the steps below.

The first level of BN consists of system components, namely generation units (G1-G11) and transmission lines (L1-L9). The parameters of the first-level nodes are determined by their FOR.

The first level of BN corresponding to the RBTS network power system consists of 19 components.

At the second level of BN, the nodes correspond to the MCs of the system. For each specified MC, a node is assigned at this level.

The parents of each specific MC node are the ones assigned to the components that create the specific MC. These parent nodes are connected to their children (MC nodes) by the arcs.

There is a logical AND relationship Between MCs and their constituent elements (parent nodes). It means if all components of an MC fails, MC will fail. Since determined

TABLE I. TABLE OF MCS OF RBTS SYSTEM AT PEAK LOAD

L9	G2G8	G7L1	G11L2	G1L4L5	G4L3L5
G1G2	G2G9	G7L2	L1L2	G2L4L5	G4L4L5
G1G4	G2G10	G8L1	L1L3	G4G8G9	G5G6L1
G1G7	G2G11	G8L2	L2L3	G4G8G10	G5G6L2
G1G8	G2G4	G9L1	L1L4	G4G8G11	G5L1L6
G1G9	G7G8	G9L2	L2L4	G4G9G10	G5L2L6
G1G10	G7G9	G10L1	L7L8	G4G9G11	G6L1L6
G1G11	G7G10	G10L2	G1L3L5	G4G10G11	G6L2L6
G2G7	G7G11	G11L1	G2L3L5	G4L3L4	G8G9G10
L3L6L8	L4L6L7	L4L6L8	L1L6L8	L3L4L5	L3L4L6
L3L6L7	L3L6L7	G9G10G11	G8G9G11	G8G9G11	

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MCs are used for the construction of the BN to evaluate LOLP based on (5), there is logical OR relationship among MCs, meaning if at least one MC is failed, the power system will fail.

$$LOLP=P\,(MC1 \cup MC2 \ \ldots \ \cup MCn) \quad (5)$$

Due to Genie & Smile software limitations, middle nodes concept are used here, which reduces the number of SF (system fault) node parameters. Fig. 3 shows the nodes at each of the BN levels corresponding to the composite power system. I_m nodes are intermediate nodes that are inserted into the model to reduce the computational burden. The parameters of these nodes will be similar to the SF nodes based on the logical OR relationship on their parent elements. So there is a logical OR relationship between I_m nodes and SF node. Thus, (CPT) of the SF node is defined based on the logical OR relation.

IV. Application of Constructed BN in Climate Change

Since the forced outage rate of system components such as transmission lines change due to climate change, the climate is one of the crucial factors affecting system reliability. In this section, the effect of this factor is evaluated by considering the climate change in the BN made in section III.B. One of the complete models discussed is the climate model. According to this model, transmission lines along with the route experience different weather conditions.

A. A transmission line with different weather conditions [11]

Over long distances, transmission lines experience different weather conditions. Different weather conditions are defined concerning a particular geographical area. Fig. 4 shows the basic model of the transmission line identifying the climatic conditions. For weather conditions, both normal and abnormal conditions are considered. S = 0 indicates normal weather, and S = 1 indicates abnormal weather. Thus a transmission line has two failure rates λ (0) and λ (1). Also, two repair rates μ (0) and μ (1) are defined for such transmission lines. Data collection schemes provide the average failure rate (λ) by the number of failures occurring per year. The relationship between λ and U and in different climates using mathematical expectation is as (6):

$$\lambda = (1-U)\lambda(0) + U\lambda(1) \quad (6)$$

In (6), U is a fraction of the time the line is in abnormal conditions and can be obtained from the probabilities of the different weather conditions that each line crosses.

The proportion of failures (F) occurring in adverse weather reflects the degree of potential failure bunching. If U and F are known, λ(0) and λ(1) can be evaluated from (7):

$$\lambda(0) = \lambda\,(1-F)/(1-U) \quad (7)$$

$$\lambda(1) = \lambda F/U$$

Since repairs are more challenging to handle in adverse weather conditions, repair time in adverse weather is expected to be longer than normal repair time.

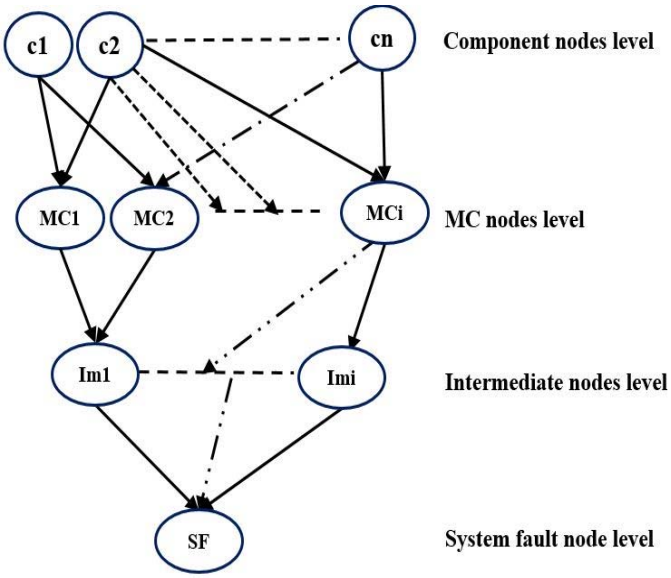


Fig. 3. Different levels of BN corresponding to RBTS system



Fig. 4. Transmission line model considering weather [11]

FOR which is used in terms of the conditions on which all the lines in normal weather, evaluated from (8):

$$FOR = \lambda(0)/(\lambda(0) + \mu(0)) \quad (8)$$

If the whole line is in adverse weather, FOR equals (9):

$$FOR = \lambda(1)/(\lambda(1) + \mu(1)) \quad (9)$$

If the line is in two or more climatic conditions, do the following: for example, when specified transmission line crosses two zones (1) and (2) when zone (1) has an adverse climate and zone (2) has a normal climate, the FOR line is calculated from (10):

$$FOR = FOR_1 + FOR_2 + FOR_1\ FOR_2 \quad (10)$$

Since in equation (10), the value of $FOR_1 \times FOR_2$ is very small, it can be ignored in practice.

B. BN Modification concerning weather

Since the effect of weather on the transmission lines is investigated in this study, it is only necessary to change the parameters of the transmission line according to different climatic conditions. For example, suppose a line is in geographic regions j and k. S_j and S_k represent the state condition of each area. Because the weather conditions of each area, effect on transmission line FOR, the nodes S_j and S_k will be the parents of that line in the Bayesian network. This is illustrated in Fig. 5.

The first step of construction the Bayesian network considering weather model is the same as mentioned in section III.B. Then the nodes of state condition weather (S_j) is added to the base BN model and the parameter of the transmissions line is modified as (11) :

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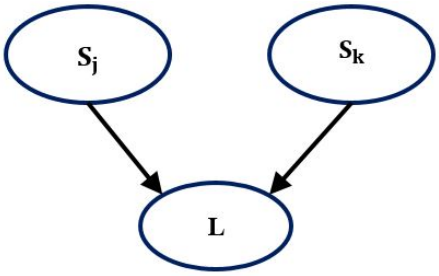


Fig. 5. Modification in node structure of L transmission line passing through two different zones

$$P_{11} = \lambda(0)/(\lambda(0) + \mu(0)) \quad (11)$$

$$P_{22} = \lambda(1)/(\lambda(1) + \mu(1))$$

$$P_{12} = \frac{\lambda(0)R}{\lambda(0)+\mu(0)} \quad + \quad P_{22} = \frac{\lambda(1)(1-R)}{\lambda(1)+(1-R)+\mu(1)}$$

$$P_{21} = \frac{\lambda(1)R}{\lambda(1)+\mu(1)} \quad + \quad P_{22} = \frac{\lambda(0)(1-R)}{\lambda(0)+(1-R)+\mu(0)}$$

Modified CPT considering weather condition is illustrated in TABLE II.

C. Case study

Simulation results are evaluated in the RBTS system considering climate change with geographical divisions. As mentioned before, RBTS has six buses, including five load points, 11 generation units, and nine transmission lines. Fig. 6 shows the climatic divisions of 3 different regions in an RBTS system. According to this classification, the percentage of transmission line in different regions is given in TABLE III. The number of region-divided weather states is 8. The related weather probabilities of the different regions are given in TABLE IV. For state 2 or 3 or 4, only one region in adverse weather, state 5 or 6, or 7 the state of two regions in adverse weather and for state 8 all three regions is in adverse weather. According to TABLE IV, the probability of different regions' climate being adverse is shown in TABLE V. The value of F, which indicates the proportion of faults occurring in adverse climates to total fault occurrences, is assumed to be 0.4. To investigate the effect of climate on the system LOLP, it is sufficient to calculate the probability P(SF = 1 | Si = 1). If we have adverse weather in several areas at the same time, for example, in areas 1 and 2, we have to calculate the probability P(S =1 | S1=1, S2=1) and represent this by R12. The results of this analysis are shown in fig. 7. As we observe, the consequence impact of inappropriate weather in Zone 3 is greater than in other areas, and LOLP is increased if Zone 3 is in an adverse state. Because a high percentage of RBTS transmission lines affected by bad weather, is in zone 3.

TABLE II. MODIFIED CPT OF A TRANSMISSION LINE CROSSING 2 DIFFERENT WEATHER STATE

S_j	Normal		Adverse	
S_k	Normal	Adverse	Normal	Adverse
Fault state	p_{11}	p_{12}	p_{21}	p_{22}
Normal state	$1-p_{11}$	$1-p_{12}$	$1-p_{21}$	$1-p_{22}$

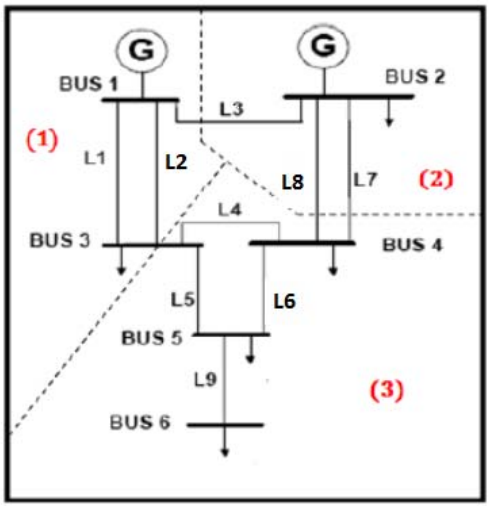


Fig. 6. Geographical three-region division [11]

TABLE III. PERCENTAGE OF TRANSMISSION LINES IN DIFFERENT AREAS

	L1	L2	L3	L4	L5	L6	L7	L8	L9
Area1	100	100	30	0	0	0	0	0	0
Area2	0	0	70	0	0	0	80	80	0
Area3	0	0	0	100	100	100	20	20	100

TABLE IV. REGION-DIVIDED WEATHER STATE PROBABILITIES

state	Adverse region	probability
1	-	0.964
2	1	0.009
3	2	0.007
4	3	0.006
5	1,2	0.005
6	1,3	0.004
7	2,3	0.004
8	1,2,3	0.001

TABLE V. PROBABILITIES OF ADVERSE WEATHER IN EACH AREA

region	Probability of adverse region
1	0.019
2	0.017
3	0.015

The results also show that the least effect on system performance error is the adverse climate of Zone 2.

Also, by constructed BN, it is possible to detect adverse weather conditions in different areas when the system fails. This is achieved by calculating the probability P (Si = 1|SF = 1) using the constructed BN. This analysis is shown in TABLE VI and Fig. 8.

TABLE VI. EFFECT PROBABILITY OF ADVERSE WEATHER CONDITION OCCURRING FAULT SYSTEM

Adverse region	Area1	Area2	Area3
P(Si=1|SF=1)	0.01892	0.04659	0.06919

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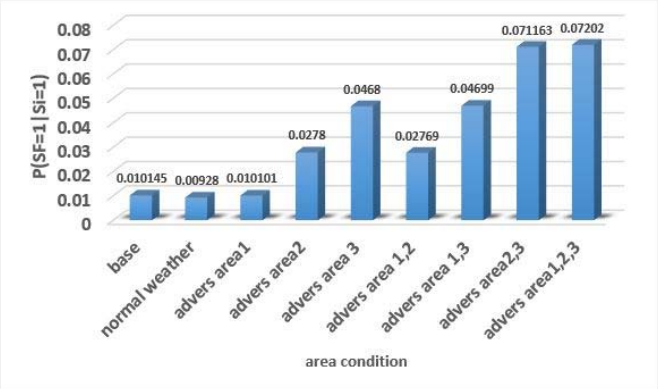


Fig. 7. Effect of weather condition on system LOLP (P(SF=1 | Si=1))

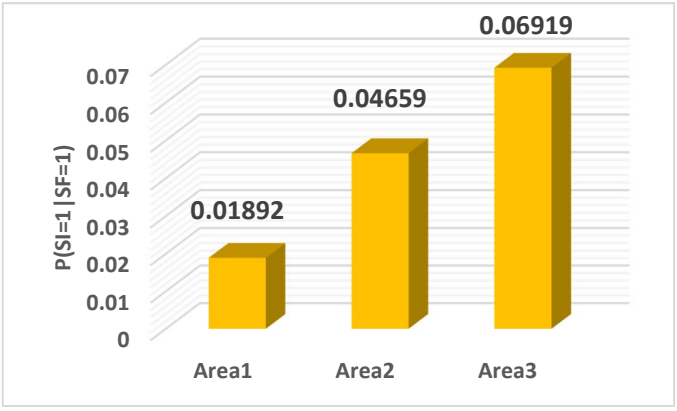


Fig. 8. Effect probability of adverse weather condition occurring system fault P(Si=1 | SF=1)

V. Conclusion

This paper proposed a framework to consider the effect of climate change on the Bayesian network model based on the MC method in order to investigate a more detailed reliability assessment of the Power system. Adverse weather may occur over a short period, but it can significantly affect the reliability of the system. In this study, one of the suitable weather models is described and used. According to this model, the whole system does not have one type of climate and, depending on the extent of the expansion, the system is geographically divided into different segments. By introducing this geographical model into the BN, the LOLP index is evaluated for each region due to adverse weather conditions and observed this model is an appropriate tool to evaluate reliability indices and other assessments. This model is more efficiency and requires less computation than other methods that assess the impact of climate on power system reliability such as Monte Carlo.

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